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A Heterogeneous Bipartite Graph Framework for Donor-Recipient Matching in Kidney Transplantation

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Abstract. This paper presents a novel graph-based framework for donor-recipient matching for organ transplantation, specifically targeting kidney transplantation. We pursue donor-recipient matching as a feature set matching problem using a graph-based approach. To represent kidney transplant clinical and operational considerations, we developed a kidney transplant graph by embedding kidney transplantation knowledge and relevant datasets to capture the complex relationships between the heterogeneous donor and recipient feature sets to provide a system for optimal donor and recipient matches. Our approach includes multi-dimensional graph edge types representing clinical considerations to provide comprehensive and optimal pairs. We propose the kidney transplant graph with a graph neural network to optimize donor-recipient compatibility predictions, enhancing decision-making in transplantation.

Keywords. Kidney Transplant, Feature Matching, Heterogeneous Bipartite Graph

1. Introduction

Organ transplantation involves matching donor and recipient clinical characteristics based on a number of factors to ensure a prolonged and functional life for the organ recipient. Kidneys filter waste from the blood, and for patients with kidney failure, dialysis, and kidney transplantation are treatment options. Whilst dialysis is the most common treatment for kidney failure, it is associated with reduced patient quality of life and a relatively lower long-term survival rate [1]. Kidney transplantation, although a more complex procedure, is an optimal treatment option for kidney failure. However, waiting times for kidney transplantation average approximately 2 to 5 years (and may be considerably longer) as the procedure depends on the availability of a donor organ. The organ allocation process involves matching the donor with the most optimal recipient to ensure a prolonged survival of the transplant [2].

Organ donor-recipient matching aims to identify the most suitable recipient for an organ donor based on immunologic matching and additional criteria, such as age, comorbidities, considerations around weight mismatch, and other factors [3], to ensure

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equity and optimal graft survival. Organ allocation for transplantation is challenging given the high demand for and limited availability of organs and the need to consider various donor and recipient factors [3]. A suboptimal donor-recipient match leads to serious consequences, such as rejection, infection, or death. Therefore, accurate matching methods are required to account for the complex interactions between the donor and recipient characteristics to achieve optimal post-transplant outcomes.

Machine learning (ML) methods have been used to effectively match donors to recipients based on feature set matching, where entities are paired based on the alignment of multiple attributes [4]. Feature set matching is widely used in computer vision, image processing, pattern recognition, and medical diagnosis matching. Feature set matching extends beyond standard classification, mapping similar entities from two distinct feature spaces whilst preserving relevant content and relationships. This complex task demands multi-dimensional matching methods that can handle heterogeneous feature spaces.

This study presents novel graph-based machine learning (ML) techniques for feature set matching to pair donors and recipients for kidney transplantation. To recommend an optimal donor and recipient match, our approach is to represent kidney transplant clinical considerations in terms of a domain-specific *kidney transplant graph* by embedding kidney transplantation knowledge and relevant datasets to capture the complex relationships between the heterogeneous donor and recipient feature sets. We present a graph neural network to perform feature set matching by assigning the most optimal recipient to an organ donor. In this paper, we present the development of the kidney transplant graph, where we model multi-dimensional graph edge types, where each edge type represents a consideration or constraint relevant to donor-recipient matching. By representing multi-dimensional graph edges, we offer a more comprehensive and optimal donor-recipient matching process that analyzes multiple relationships between donor and recipient features. Figure 1 illustrates the proposed donor-recipient matching pipeline—using graph representation, we embed donor and recipient features into a shared feature space, and graph neural networks bring compatible pairs closer together. Our approach demonstrates the effectiveness of machine learning for donor-recipient matching.

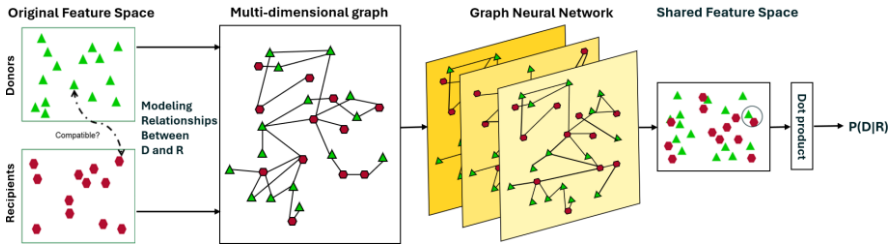


Figure 1. Donor-recipient matching pipeline through graph embedding.

2. Background

Donor-recipient matching employs two primary techniques. Conventional metrics based on standard guidelines use predefined rules, criteria, and equations to measure the similarity between feature sets and find the best match [5]. ML methods are used as classifiers to identify complex relationships between donors and recipients [7]. Don Yoo et al. [7] used ensemble learning techniques including survival decision trees and random forests to predict long-term survival. Naqvi et al. [1] adopted a comprehensive approach

to optimize donor-recipient matching to predict graft survival over various time intervals following transplantation. While prevalent donor-recipient matching approaches provide valuable information, they do not always capture non-linear relationships between the variables, especially as the dimensionality of the data increases.

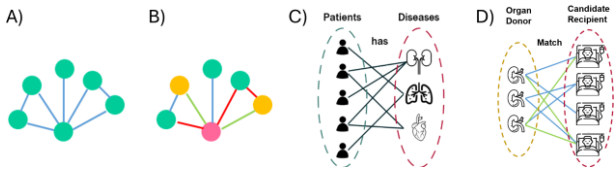


Figure 2. A) Homogeneous graph, B) Heterogeneous graph, C) Bipartite graph showing patient-disease relationships, D) Organ donor-candidate recipient interactions.

Graph-based techniques offer a state-of-the-art approach for matching distinct feature spaces, such as donor and recipient features. A graph G consists of vertices (V) and edges (E) to represent entities and their relationships. Homogeneous graphs consist of nodes and edges of a single type, whereas heterogeneous graphs include multiple nodes and edges, offering a more information-rich representation to capture diverse data aspects and dimensions. A bipartite graph, Figure 2, is a heterogeneous graph where nodes are divided into two sets representing inter-domain interactions.

3. Methodology

This study used data from the Scientific Registry of Transplant Recipients (SRTR). The SRTR data system includes data on all donor, wait-listed candidates, and transplant recipients in the US, submitted by the members of the Organ Procurement and Transplantation Network (OPTN). The Health Resources and Services Administration (HRSA), U.S. Department of Health and Human Services provides oversight to the activities of the OPTN and SRTR contractors. The data reported here have been supplied by the Hennepin Healthcare Research Institute (HHRI) as the contractor for the Scientific Registry of Transplant Recipients (SRTR). The interpretation and reporting of these data are the responsibility of the author(s) and in no way should be seen as an official policy of or interpretation by the SRTR or the U.S. Government.

3.1. Compatibility Edges

Compatibility edges link compatible donor-recipient pairs within the dataset based on their characteristics and post-transplant survival days. A pair is considered compatible if they completed a kidney transplant and met a survival threshold, such as 5 or 8 years post-transplant survival, balancing outcomes with follow-up. These compatibility edges create one-to-one connections, resulting in a sparse graph where most nodes have few or no neighbors. To address this sparsity, we introduced additional edges to enrich the graph.

3.2. Knowledge-Driven Edges

Knowledge-driven edges enrich the graph by incorporating prior knowledge and domain expertise. We utilized the allocation strategy proposed by Baskin-Bey et al., which calculates the RRS and Deceased Donor Score (DDS) based on key clinical variables,

including the history of diabetes, history of angina, and time on dialysis therapy, and age. We used 12 identified donor-recipient matching scenarios based on specific combinations of these variables [6] and derived DDS using a more recent equation for calculating final creatinine clearance. The DDS is a scoring system that combines various donor-related variables into a cumulative score ranging from 0 to 39 points. Donor kidneys are classified into four quality grades: grade A (0-9 points), grade B (10-19 points), grade C (20-29 points), and grade D (30-39 points). Key variables for the DDS include donor age, hypertension history, final creatinine clearance (replaced by estimated glomerular filtration rate), HLA mismatch, and cause of death. We employed the 2021 CKD-EPI creatinine equation to estimate the glomerular filtration rate (GFR) in mL/min/1.73 m², kidney function indicators varying by age and sex, as shown in Eq. (1).

$$eGFR = 142 \times min\left(\frac{S_{cr}}{k}, 1\right)^{\alpha} \times max\left(\frac{S_{cr}}{k}, 1\right)^{-1.200} \times 0.9938^{age} \times [1.012, if\ female] \quad (1)$$

where Scr is the serum creatinine, k is 0.7 for females and 0.9 for males, and α is -0.241 for females and -0.302 for males [8]. Donor kidneys are classified as grades A, B, C, and D based on the DDS, and are allocated to recipients in groups RG1, RG2, RG3, and RG4, respectively, according to their RRS classifications [6].

3.3. Blood Type and Cluster-Based Edges

Blood type edges allow better generalization in cases with multiple compatibility factors. For instance, donors with blood type O can donate to all recipient blood types, while type A can donate to A and AB, type B to B and AB, and type AB only to AB recipients. Cluster-based edges enhance transplant match detection by forming clusters with the highest success likelihood, representing underlying relationships for more accurate matches. This method improves model generalization by focusing on broader patterns, reducing noise, and prioritizing relevant connections.

3.4. Time-Based Edges (Temporal Urgency)

Time-based edges capture the urgency in decision-making by connecting donor-recipient pairs within transplant time window, considering advanced kidney disease, organ viability, and waitlist time. These edges add dynamism to the graph, enabling real-time decision-making and prioritization. While adding diverse edge types can enrich the graph, overloading it with edges may lead to a fully connected graph, reducing its utility.

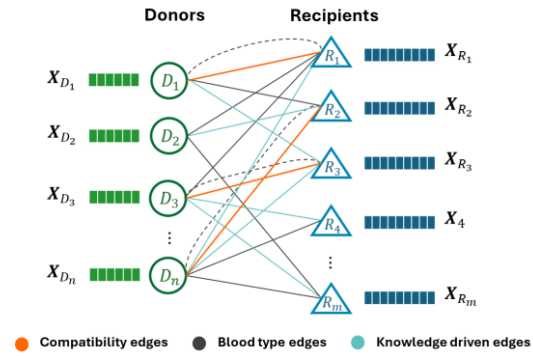


Figure 3. The proposed heterogeneous bipartite multigraph capturing the relation of donors and recipients

4. Results

We developed a kidney transplant graph from the SRTR dataset, transforming the tabular data into a donor-recipient heterogeneous graph $G = (V, E)$, consisting of two distinct types of nodes: 5,000 donor nodes (V_D) and 5,000 recipient nodes (V_R). Each donor node $v_D \in V_D$ has a feature vector $X_D \in \mathbb{R}^{26}$ containing 26 attributes related to donor characteristics, while each recipient node $v_R \in V_R$ has a feature vector $X_R \in \mathbb{R}^{31}$ with 31 attributes related to recipient characteristics. The total node set is represented as $V = V_D \cup V_R$ with $V_D \cap V_R = \emptyset$. The final graph, by connecting each donor (D_1 to D_n) to their corresponding recipients (R_1 to R_m) using compatibility, knowledge-driven, and blood type edges, resulted in a bipartite graph with 18M edges. Figure 3 highlights the multi-dimensional donor-recipient relationships as graph edges.

5. Discussion and Conclusions

This paper presents a novel graph-based framework for donor-recipient matching for kidney transplantation. By representing donors and recipients as interconnected entities in a heterogeneous bipartite graph, our framework captures critical relationships and constraints often missed by traditional methods. This graph-based representation of donor-recipient relationships supports the application of advanced graph neural networks (GNN) to aggregate and process such complex relational data. By representing donor-recipient samples as a connected graph, information from observed donor-recipient pairs can be transferred to other donor and recipient samples to further investigate non-observed relationships between donor and recipient features. While this is a promising approach, it introduces the challenges of defining meaningful edges for non-graph problems such as kidney transplantation, highlighting the importance of edge design. Future research will explore different edge definitions in a GNN-based prediction setup. This ongoing work aims to provide clinicians with a tool that illustrates the abstract relationships between donors and recipients for different transplant scenarios, thereby helping to improve transplantation outcomes. This proposed framework can be extended to other organ transplant studies.

References

[1] Naqvi SA, et al. Predicting kidney graft survival using ML methods: prediction model development and feature significance analysis study. Journal of Medical Internet Research. 2021 Aug 27;23(8):e26843.
[2] Majouni S, et al. Predicting Urgent Dialysis at Ambulance Transport to the Emergency Department Using ML Methods. In MEDINFO 2023—The Future Is Accessible 2024 (pp. 891-895). IOS Press.
[3] Vinson AJ, Tennankore KK, Kiberd BA. Donor-Recipient Matching in Kidney Transplant: We're Not There Yet. In Transplantation Proceedings 2021 Jul 1 (Vol. 53, No. 6, pp. 1909-1914). Elsevier.
[4] Baumberg A. Reliable feature matching across widely separated views. In: Proceedings IEEE Conf Comput Vis Pattern Recognit (CVPR); 2000 Jun 15; vol. 1. New York: IEEE; 2000. p. 774-81.
[5] Schwantes IR, Axelrod DA. Technology-enabled care and artificial intelligence in kidney transplantation. Current transplantation reports. 2021 Sep;8:235-40.
[6] Baskin-Bey ES, Kremers W, Nyberg SL. A recipient risk score for deceased donor renal allocation. American journal of kidney diseases. 2007 Feb 1;49(2):284-93.
[7] Yoo KD, Noh J, et al. A machine learning approach using survival statistics to predict graft survival in kidney transplant recipients: a multicenter cohort study. Scientific reports. 2017 Aug 21;7(1):1-2.
[8] Inker LA, Eneanya ND,et al. New creatinine-and cystatin C–based equations to estimate GFR without race. New England Journal of Medicine. 2021 Nov 4;385(19):1737-49.